

Answer Key — Biology Grade 11

Sub-Strand 2.2: Growth and Development in Plants • Lesson 3: Seed Dormancy: Why Do Healthy-Looking Seeds Not Germinate?

Teacher reference only. Do not project or hand out.

Quick marking grid

Q1	Q2	Q3	Q4	Q5	Q6	Q7	Q8
A	A	C	C	C	B	B	A

USE AFTER: PREDICT PHASE — Predict: Think-Pair-Share on the data card (clarifying 'viable' and 'dormant' after predictions)

Q1. Mr. Wafula planted Rosecoco beans and some healthy-looking seeds did not germinate even though the rain, soil and planting day were the same. Which statement best describes a dormant seed?

- A. A living seed that fails to germinate even when conditions are favourable**
- B. A seed that has been eaten by pests in the soil
- C. A dead seed whose embryo has rotted
- D. A seed that fails because it was planted in waterlogged soil

Correct answer: A. Dormancy means a viable (alive) seed does not germinate even in favourable conditions. Dead seeds, pest damage and waterlogging are different causes of failure.

USE AFTER: OBSERVE PHASE — Part B: Sort the non-germinating seeds (data card table for beans)

Q2. Of 100 Rosecoco bean seeds planted, 20 did not germinate. The reasons were: 8 had a hard impermeable seed coat, 6 were in a waterlogged patch, 4 were old seed with dead embryos, and 2 were eaten by pests. How many of the 20 non-germinating beans failed for reasons OTHER than dormancy?

- A. 12**
- B. 14
- C. 8
- D. 6

Correct answer: A. Only the 8 hard-coated seeds were dormant. The others are $6 + 4 + 2 = 12$, which is also $20 - 8 = 12$.

USE AFTER: OBSERVE PHASE — Part B: Sort the non-germinating seeds (comparing maize with 0 dormant and beans with 8 dormant)

Q3. In Mr. Wafula's field, 12 maize seeds did not germinate: 5 were eaten or rotted by fungi or pests, 4 were old seeds with dead embryos, and 3 were in a waterlogged patch. Among the 20 non-germinating beans, 8 had a hard impermeable seed coat. Which conclusion is supported by this data?

- A. Maize seeds can never be dormant
- B. Dormancy caused all the bean failures but none of the maize failures
- C. Dormancy explained 8 of the 20 bean failures and none of the 12 maize failures in this field**
- D. The hard seed coat caused most of the maize failures

Correct answer: C. The table shows 8 dormant beans out of 20 failures and 0 dormant maize out of 12. The data shows that not germinating has several causes, and it cannot show that maize never has dormancy.

USE AFTER: OBSERVE PHASE — Part C: Field result (7 of 8 germinated after scarification)

Q4. The 8 hard-coated bean seeds were scarified (nicked) and replanted. By Day 7, 7 of the 8 had germinated. What percentage of the 8 seeds germinated?

- A. 70%
- B. 7.8%
- C. 87.5%**
- D. 12.5%

Correct answer: C. Percentage = $7 \div 8 \times 100 = 87.5\%$. The 12.5% option is the share that did not germinate.

USE AFTER: EXPLAIN PHASE — Explain: CER explanation of why 8 healthy-looking seeds did not germinate and why 7 of 8 germinated after scarification

Q5. Why did scarifying (nicking or rubbing) the hard seed coat of the dormant bean seeds help them germinate?

- A. It added food to the seed so the embryo could grow
- B. It made the embryo larger before planting
- C. It made a small opening in the impermeable testa so water and oxygen could reach the embryo**
- D. It killed the pests living inside the seed

Correct answer: C. A hard impermeable testa blocks water and oxygen. Scarification damages the testa gently so that water and oxygen can enter and the living embryo can start to grow.

USE AFTER: EXPLAIN PHASE — Explain: matching each cause of dormancy to a way of breaking it

Q6. A farmer finds that some seeds stay dormant because of a chemical inhibitor, abscisic acid, in the seed. Which way of breaking dormancy matches this cause best?

- A. Planting the seeds in a waterlogged patch
- B. Soaking the seeds in water for some hours so the inhibitor is washed out**
- C. Rubbing the seed coat with sandpaper
- D. Storing the seeds in sunlight and adding fertiliser

Correct answer: B. The way of breaking dormancy must match the cause. A chemical inhibitor can be washed out (leached) by water, whereas sandpaper is for a hard testa.

USE AFTER: DQB PHASE — Driving Question Board: moving 'Why do healthy-looking seeds not germinate?' to 'Partly answered' and the exit ticket

Q7. A learner says, 'A seed that does not germinate must be dead.' Which group of non-germinating bean seeds from Mr. Wafula's field shows that this idea is wrong?

- A. The 4 old seeds with dead embryos
- B. The 8 hard-coated seeds, of which 7 germinated after scarification**
- C. The 5 maize seeds eaten or rotted by fungi
- D. The 2 seeds eaten by pests

Correct answer: B. The hard-coated seeds were alive but dormant, and 7 of the 8 germinated once the testa was nicked. The other groups were dead, eaten or rotted.

USE AFTER: END OF LESSON — Explain: significance of dormancy to the plant and to the farmer

Q8. Why can dormancy be useful to a wild plant but a problem for a farmer like Mr. Wafula?

- A. Dormancy helps seeds survive dry or cold seasons and germinate over time, but the farmer wants all seeds to germinate together for a uniform stand**
- B. Dormant seeds always die, so wild plants make more seeds, but the farmer loses money
- C. Dormancy stops pests from eating seeds in the wild but not in a farm
- D. Dormancy makes seeds grow faster, which wild plants need but farmers do not

Correct answer: A. For wild plants, dormancy keeps seeds alive through unfavourable seasons and spreads germination over time. A farmer needs even germination, so scarification is used to break dormancy.

